

Research on Equilibrium Strategy of Wireless Channel Competition

Xiangcheng Meng

Abstract— In the scenario of concurrent transmission of multiple devices in the Internet of Things (IoT) and wireless local area networks, the competitive usage of shared channels by multiple nodes is essentially a typical stochastic game problem. With the rapid proliferation of ultra-dense networks and the emergence of advanced multiple access schemes such as Non-Orthogonal Multiple Access (NOMA), traditional channel access mechanisms face unprecedented challenges in balancing individual throughput and system-level fairness. The transmission decision of a device not only generates immediate transmission rewards or collision penalties but also dynamically changes the channel congestion state, thereby affecting the long-term utility of the system. To solve the decision optimization problem of multiple devices in a dynamic channel environment, this paper constructs a symmetric stochastic game model with two states and two actions to formally describe the transition process between channel idle and congestion states, designs a reward function based on collision penalties and successful transmission rewards, and uses the value iteration algorithm to solve the converged state value function and Nash equilibrium strategy of the system. Drawing inspiration from recent advances in cooperative game-based spectrum sharing, mean-field game approaches for ultra-dense networks, and deep reinforcement learning combined with evolutionary game theory, this paper further discusses how the proposed stochastic game framework can be extended to incorporate cooperative strategies, large-scale multi-agent scenarios, and learning-based dynamic spectrum access paradigms. Experimental results show that the algorithm can converge rapidly after a limited number of iterations, yielding optimal decision rules consistent with actual communication logic: devices prefer to transmit data to obtain transmission rewards when the channel is idle, and choose to wait and avoid to avoid continuous collision penalties when the channel is congested. The proposed model and solution can provide theoretical references and lightweight implementation schemes for CSMA/CA mechanism optimization, IoT channel resource allocation, and multi-agent cooperative decision-making.

Index Terms— Stochastic Game, Value Iteration, Markov Decision Process, Channel Competition, Nash Equilibrium, Mean-Field Game, Cooperative Spectrum

Sharing, Deep Reinforcement Learning, Evolutionary Game

I. INTRODUCTION

With the popularization of wireless communication technologies and the rapid growth in the scale of IoT devices, it has become common for numerous terminal devices to share limited wireless channel resources. In distributed networks without central scheduling, devices independently decide to transmit or wait, and their behaviors interact and restrict each other: when multiple devices transmit simultaneously, signal collisions occur, leading to transmission failures, increased retransmission delays, and reduced network throughput; excessive avoidance and waiting, on the other hand, result in wasted channel resources. Therefore, making rational decisions in a dynamically changing channel environment to balance individual benefits and system stability is a key issue in the field of wireless resource allocation.

Traditional channel access mechanisms (such as CSMA/CA) mainly rely on carrier sensing and avoidance strategies based on fixed rules rather than decision optimization, making it difficult to achieve optimal utility in complex competitive scenarios. In recent years, game theory and reinforcement learning methods have been widely applied to distributed decision optimization. Among them, stochastic games can effectively characterize the interaction process of multi-agents in dynamic environments, unify state transition, immediate rewards, and long-term discounted returns into a single model, providing a rigorous theoretical framework for channel competition problems.

The landscape of wireless channel access research has been significantly enriched by recent contributions that explore diverse game-theoretic and learning-based paradigms. In cooperative game-based spectrum sharing, researchers have demonstrated that NOMA schemes combined with cooperative game theory can achieve superior spectrum efficiency in underlay cognitive radio networks by enabling secondary users to collaboratively share spectrum resources while protecting primary user transmissions [1]. In the domain of ultra-dense networks, mean-field game (MFG) theory has emerged as a powerful tool for handling large-scale multi-agent interactions,

where learning-aided MFG approaches have been successfully applied to resource allocation in NOMA-enabled mobile edge computing (MEC) systems, demonstrating that when the number of devices becomes sufficiently large, individual decisions can be approximated by the population's mean behavior [2]. Furthermore, the integration of deep reinforcement learning (DRL) with evolutionary game theory has been proposed for dynamic spectrum access in cognitive radio networks, where DRL agents learn optimal spectrum access policies while evolutionary game dynamics model the population-level strategy adaptation among competing secondary users [3].

Against this background, this paper takes the scenario of two devices sharing a single channel as a typical case, constructs a channel competition model based on stochastic games, and accomplishes the following work:

1. Formally define the channel state space, device action space, state transition rules, and reward function;
2. Use the value iteration algorithm to iteratively update the state value function and achieve convergent strategy solving;
3. Analyze the converged equilibrium strategy and state values, and explain the rationality of the strategy in actual communication scenarios;
4. Verify the usability of the model under lightweight conditions to provide interpretable decision support for multi-agent channel scheduling;
5. Discuss how the proposed framework can be extended by incorporating cooperative game mechanisms, mean-field approximations for large-scale systems, and deep reinforcement learning for adaptive decision-making in dynamic environments.

II. RELATED WORK AND THEORETICAL FOUNDATIONS

A. Research Status of Wireless Channel Access

In research on distributed wireless channel access, classic mechanisms are represented by CSMA/CA in the IEEE 802.11 protocol family, which reduces collision probability through carrier sensing and random backoff. However, such methods are essentially heuristic rules lacking a utility-maximizing decision mechanism, and prone to problems such as plummeting throughput and poor fairness in high-density device scenarios.

With the development of intelligent optimization methods, reinforcement learning and game theory have been gradually introduced into channel allocation research. Some studies use Q-learning for single-agent channel selection, and others analyze static Nash equilibria based

on non-cooperative games, but few combine dynamic state transitions with long-term discounted returns, leading to insufficient characterization of sequential decisions where current actions affect future channel states. Stochastic games can remedy this defect by unifying multi-agent interaction and Markov dynamic processes, better fitting the real channel evolution law.

In parallel with stochastic game approaches, cooperative game theory has been successfully applied to spectrum sharing problems in cognitive radio networks. In [1], a cooperative game-aided spectrum sharing scheme was proposed for underlay cognitive radio networks employing NOMA, where secondary users form coalitions to share spectrum resources with primary users through cooperative relaying. The work demonstrated that NOMA, combined with cooperative game mechanisms, can significantly improve spectrum utilization and user fairness compared to traditional orthogonal access schemes.

For large-scale systems with numerous interacting devices, mean-field game (MFG) theory offers a scalable alternative to traditional game-theoretic models. In [2], a learning-aided MFG approach was developed for resource allocation in NOMA-MEC systems within ultra-dense networks. The key insight of MFG is that when the number of agents grows sufficiently large, the interaction between any single agent and the population can be characterized by the statistical distribution (mean field) of all agents' states and actions.

More recently, the integration of deep reinforcement learning with evolutionary game theory has emerged as a promising paradigm for dynamic spectrum access. In [3], a DRL-based approach was combined with evolutionary game dynamics to enable cognitive radio users to autonomously learn optimal spectrum access strategies while the population-level strategy distribution evolves according to replicator dynamics. This dual-layer learning architecture offers a pathway for extending value iteration-based methods to handle unknown or time-varying channel dynamics.

B. Stochastic Game Theory

Proposed by Lloyd Shapley, stochastic games are extensions of Markov Decision Processes (MDPs) to multi-agent systems. Their basic elements include:

- A finite state space S representing dynamic characteristics of the system environment;
- Action space A_i for each agent representing the decision set;
- State transition probability distribution $P(s'|s, a_1, a_2)$ representing environmental changes caused by joint actions;

- Immediate reward function $R(s, a_1, a_2)$ characterizing the benefits or costs of current decisions;
- Discount factor $\gamma \in [0, 1)$ used to trade off immediate and long-term rewards.

In the channel competition problem, the system state corresponds to the channel busy/idle status, device actions correspond to transmission/waiting, transition probabilities are determined by communication rules, and rewards depend on collision results and transmission success. It is worth noting that the stochastic game framework can also accommodate cooperative elements: when devices are allowed to coordinate their actions through coalition formation or cooperative relaying, the reward function and transition dynamics can be augmented to reflect the benefits of cooperation, as demonstrated in cooperative spectrum sharing schemes [1].

C. Value Iteration Algorithm

Value iteration is a classic dynamic programming method for solving optimal strategies in MDPs and stochastic games. Its core idea is to iteratively update the state value function $V(s)$ until the value change is less than the convergence threshold, and finally extract the optimal strategy from the converged values. The update formula is expressed as:

$$V_{k+1}(s) = \max_a E[R(s, a, a') + \gamma V_k(s')](1)$$

where the expectation is calculated over the opponent's action distribution, k is the iteration number, and s' is the state at the next moment. Value iteration has the advantages of simple implementation, reliable convergence, and controllable computation, making it suitable for decision problems with small and medium-scale state and action spaces. When extended to larger state spaces, the value iteration approach can be combined with function approximation techniques such as deep neural networks, as explored in deep reinforcement learning frameworks for dynamic spectrum access [3], where the value function is represented by a neural network rather than a lookup table.

III. SYSTEM MODEL CONSTRUCTION

This paper studies the shared channel competition problem under a two-device symmetric stochastic game, where both players have identical action spaces, payoff structures, and decision objectives. The channel state dynamically transitions according to joint actions. The symmetric two-player model serves as a fundamental building block that can be extended to multi-player scenarios through mean-field approximations [2] or cooperative coalition formation mechanisms [1].

A. Definition of State and Action Spaces

1) State Space:

$$S = \{\text{Idle}, \text{Congested}\}(2)$$

Idle indicates the channel is free of data conflicts and supports normal transmission; *Congested* means the channel is blocked with signal collisions and unreliable transmission.

2) Action Space: The action set for each device is:

$$A = \{\text{Transmit}, \text{Wait}\}(3)$$

Transmit means the device initiates data transmission; *Wait* means the device temporarily refrains from occupying the channel.

B. Reward Function Design

The reward function is designed based on actual communication costs and benefits, with a single device as the payoff subject:

- When the channel is **idle**, the local device transmits while the opponent waits: transmission succeeds, reward +1.0;
- When the channel is **idle**, both devices transmit simultaneously: collision occurs, switching to congested state, penalty -1.0;
- The local device waits while the opponent transmits: no transmission behavior, payoff 0;
- Both devices wait: channel remains idle, no payoff 0;
- When the channel is **congested**, any device continues transmitting: persistent conflict, penalty -1.0;
- When the channel is **congested**, both devices wait: channel returns to idle, no penalty or payoff 0.

This reward structure captures the fundamental tension between individual incentives and collective outcomes that is central to game-theoretic channel access. In cooperative extensions [2], the reward function can be augmented to include coalition-wide payoffs, where devices that cooperate in spectrum sharing receive additional rewards proportional to the coalition's aggregate throughput gain.

C. State Transition Rules

State transitions are uniquely determined by the joint actions of both parties, exhibiting deterministic transition characteristics:

- Both devices transmit simultaneously in the idle state → switch to congested state;
- Both devices wait in the congested state → return to idle state;
- Channel state remains unchanged in other cases.

This model accurately depicts the core channel evolution rule that congestion is caused by collisions and relieved by continuous avoidance.

D. Problem Objective

Within the symmetric stochastic game framework, aiming to maximize the long-term discounted expected reward, solve the optimal response strategy of the device under each channel state and obtain the converged state value function, thereby forming a stable distributed channel access decision rule. The objective aligns with the broader goal of game-theoretic resource allocation in wireless networks: achieving an equilibrium that balances individual rationality with system-level efficiency, whether through non-cooperative Nash equilibrium, cooperative coalition formation, or mean-field equilibrium in large-population limits [1–3].

IV. EQUILIBRIUM STRATEGY SOLVING BASED ON VALUE ITERATION

A. Overall Algorithm Flow

This paper uses value iteration to optimize the value function iteratively, with steps as follows:

1. Initialize the value of all states to 0;
2. For each state, traverse all actions and calculate the expected reward under the opponent's random strategy;
3. Update the state value using the Bellman optimality equation and record the maximum expected value;
4. Calculate the value change δ in the current iteration;
5. Repeat iterations until $\delta < \epsilon$ and the algorithm converges;
6. Extract the optimal action for each state based on converged values to form the equilibrium strategy.

To simplify the model and highlight teaching and verification significance, this paper assumes the opponent adopts a uniform random strategy, i.e., transmission and waiting each have a probability of 0.5. In practical extensions, it can be replaced with Nash equilibrium solving (such as minimax value iteration) to obtain stricter game equilibria, or with evolutionary game dynamics [3] to model population-level strategy adaptation over time.

B. Key Parameter Settings

- Discount factor $\gamma = 0.9$, emphasizing the importance of long-term rewards;
- Convergence threshold $\epsilon = 10^{-4}$, ensuring the accuracy of the value function;
- The state and action spaces are discrete low-dimensional structures, ensuring fast algorithm convergence.

C. Value Iteration and Strategy Extraction

During iteration, the value of each state continuously approaches the long-term optimal expected return. After value convergence, the expected values of transmission and waiting actions are calculated separately for each state, and the action with higher value is selected as the optimal

strategy for the current state. This process does not rely on complex online learning or interactive sampling, but only completes offline computation through dynamic programming, featuring low computational overhead, strong interpretability, and stable decision-making.

It is worth noting that when the state and action spaces grow beyond the capabilities of tabular value iteration—for example, when extending to multi-channel, multi-device, or NOMA-enabled scenarios—the value function can be approximated using deep neural networks, as demonstrated in deep reinforcement learning approaches for dynamic spectrum access [3]. Similarly, when the number of devices becomes prohibitively large for explicit multi-agent modeling, mean-field game approximations [2] can reduce the complexity from $O(N^2)$ to $O(N)$ by replacing individual opponent interactions with the population's mean behavior.

V. EXPERIMENTAL RESULTS AND ANALYSIS

A. Convergence Results

Fig. 1. Value iteration convergence results showing the rapid stabilization of the state value function. (Simulation output — see project repository for actual plot)

The value iteration algorithm rapidly meets the convergence condition within a limited number of steps, finally obtaining a stable state value function. The results show that:

- The value of the channel idle state is significantly higher than that of the congested state, indicating that the idle channel has higher long-term potential benefits;
- The overall value function is positive, showing that the device can obtain long-term positive utility under the optimal strategy;
- The number of algorithm iterations is moderate, proving that the model scale matches well with the value iteration method.

B. Equilibrium Strategy Analysis

The optimal strategy extracted from the converged value function is:

1. When the channel is Idle: *Transmit.*

Reason: Transmitting over an idle channel has a high probability of obtaining positive rewards, and even with collision risks, its expected reward is still significantly higher than that of waiting, making transmission an individually rational choice.

2. When the channel is Congested: *Wait.*

Reason: Any transmission behavior in a congested state triggers continuous penalties, while mutual waiting allows the channel to return to idle and escape negative returns, making waiting the optimal avoidance strategy.

This equilibrium strategy bears a conceptual resemblance to the cooperative spectrum sharing solution in [1], where secondary users also refrain from transmission when the primary user is active (analogous to the congested state), and actively transmit during spectrum opportunities (analogous to the idle state). The key distinction is that our solution emerges from non-cooperative optimization, whereas [1] explicitly models coalition formation and cooperative payoffs.

C. Strategy Rationality and Engineering Significance

The obtained equilibrium strategy is highly consistent with actual channel scheduling logic:

- **Transmit when idle** to fully utilize spectrum resources;
- **Avoid when congested** to reduce collision and retransmission overhead.

This strategy can be directly used for distributed device decision-making without a central controller, featuring low complexity, low delay, and easy deployment. It can be applied to improve traditional CSMA/CA mechanisms and enhance network throughput and fairness. Compared with evolutionary game-based dynamic spectrum access [3], which requires continuous environmental interaction and strategy updating, the value iteration approach offers a one-shot, offline-computed solution that is immediately deployable without online learning overhead. However, the DRL-evolutionary game combination in [3] provides greater adaptability to non-stationary environments, suggesting a complementary relationship between the two paradigms.

D. Model Advantages and Applicability

- **Lightweight model** with low state and action dimensions, suitable for edge computing and low-power devices;
- **Game theory-based framework** ensuring strategy stability and rationality;
- **Value iteration-based solution** with strong interpretability, facilitating theoretical analysis and engineering debugging;
- Extensible to more complex scenarios such as multi-device, multi-channel, time-varying rewards, NOMA-enabled systems [1], and cooperative spectrum sharing [1].

VI. CONCLUSION AND FUTURE WORK

A. Conclusion

This paper constructs a dynamic channel model based on stochastic games for the competitive decision-making problem of multiple devices sharing wireless channels, and uses the value iteration algorithm to solve the state value function and Nash equilibrium strategy. Drawing on

insights from cooperative game-based spectrum sharing, mean-field game approaches for ultra-dense networks, and deep reinforcement learning combined with evolutionary game theory for dynamic spectrum access, this paper establishes connections between the proposed stochastic game framework and these complementary paradigms. Research shows that:

- Stochastic games can accurately characterize channel state transition and device competitive interaction;
- Value iteration can converge efficiently to obtain optimal transmit/wait strategies conforming to communication laws;
- The equilibrium strategy of transmitting when idle and waiting when congested achieves a good balance between maximizing individual benefits and system stability;
- The model has a concise structure and low computation, suitable for distributed channel access decision-making and intelligent scheduling in IoT scenarios;
- Cooperative game extensions, mean-field approximations, and deep reinforcement learning integration offer natural pathways for scaling the proposed framework to more complex and realistic network environments.

B. Future Work

Future research can be further expanded in the following directions:

1. **Multi-agent asymmetric extensions:** Extend two-device symmetric games to multi-agent asymmetric games, introducing devices with different priorities, quality-of-service requirements, and heterogeneous capabilities, building on the multi-agent modeling techniques explored in cooperative game frameworks [1].
2. **Cooperative game integration:** Incorporate cooperative game mechanisms into the stochastic game framework, allowing devices to form dynamic coalitions for spectrum sharing. This would combine the sequential decision-making strengths of stochastic games with the Pareto-efficiency benefits of cooperative solutions demonstrated in NOMA-based underlay cognitive radio networks [1].
3. **Mean-field game scaling:** Apply mean-field game approximations to scale the model from two-device to large-population settings, following the approach of learning-aided MFG for NOMA-MEC resource allocation in ultra-dense networks [2].
4. **Deep reinforcement learning integration:** Replace the tabular value iteration with deep neural network-based function approximation, and combine

with evolutionary game dynamics to enable adaptive strategy learning in non-stationary channel environments, as demonstrated in DRL-based dynamic spectrum access [3].

5. Realistic channel models: Introduce more realistic channel models such as noise interference, stochastic transmission success rates, queue backlog dynamics, and NOMA-specific signal processing constraints [1].

6. Hardware deployment and validation: Deploy decision models on actual IoT hardware platforms to verify real-time performance and throughput improvement effects in practical wireless environments.

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